Yash Vardhan

cs25mtech14016@iith.ac.in | eryash0401@gmail.com | +91 6203523212 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.46
Mount Assisi School	Class XII (ISC)	2018	89.40%
Mount Assisi School	Class X (ICSE)	2016	92.33%

PROFESSIONAL EXPERIENCE

IBM | RDBMS Training | Link

- **Engineered and optimized** complex **SQL** and **PL/SQL** queries in **Oracle DB** to resolve real-world relational data scenarios, **improving retrieval accuracy** across hands-on training modules.
- **Automated** Unix-based database and file-system workflows on Ubuntu, streamlining routine operations and cutting repetitive manual administration effort.

PROJECTS

Federated Multimodal Unlearning: Enabling the Right to be Forgotten in VLMs under Decentralized Data | Ongoing | Repo

- **Established** a federated multimodal unlearning testbed for medical VLMs, fine-tuning **InternVL3-7B** with frozen backbone with **traninable LoRA and InternVL vision adapter** across 3 non-IID hospital clients on SLAKE; **formulated a 7-metric evaluation protocol** spanning forgetting, retention, modality selectivity, cross-client harm, and durability.
- **Benchmarked 6 SOTA methods** (GA, NPO, Ferrari, FUSED, SIU, MMUnlearner), finding that **all relapse within 10 FedAvg rounds** with forgetting efficacy rebounding to **0.76-0.86**; proposed a durable framework combining **DP-noised Fisher masks, calibrated masked optimization, and a server-minted Forget Certificate** to enforce forgetting across future federated updates.

DiFUSED: Data-Efficient Federated Unlearning via Distribution-Matched Distillation with Selective Sparse Adapters | Report | Repo

- **Devised DiFUSED**, a distillation stage for reversible federated unlearning that freezes the federated backbone, strips its classifier head, and optimizes raw pixels against an MMD loss over L2-normalized per-class mean embeddings, synthesizing **5 class-balanced images per class per client** as proxies for real retain data during sparse-adapter training.
- **Reduced 56K real retain images to 1.9K synthetic proxies (30× compression)**, cutting adapter-training wall-clock by **59-62% on FashionMNIST/LeNet** and **21-22% on CIFAR-10/ResNet-18** across 50-client federations; retain accuracy improved **0.7-4.5 pp** while membership-inference accuracy remained **0.66**.

MoneyFlow | Report | Repo

- **Architected** a secure PHP 8.2 banking platform with strict **Content-Security-Policy and CSRF enforcement**, incorporating framework-free **Argon2id, SHA-256 token sessions bound to IP/User-Agent, AES-256-GCM transaction-intent binding, TOCTOU-safe atomic rate limiting**, and least-privilege MySQL access; deployed via Docker Compose with non-root Apache and **TLS 1.3**.
- **Led incident response** during a **7-day adversarial war game involving 15+ attacking teams**, triaging and reproducing **13 live attack reports** spanning SQLi, session fixation, malicious file upload, brute force, and SSL stripping; performed root-cause analysis and remediation with **zero successful compromises**.

Privacy Preserving Synthetic Data Generation | Report | Repo

- **Engineered** a privacy-preserving synthetic data generator using a -VAE with mixed **MSE + cross-entropy + KL objectives** and posterior sampling, coupled with a three-stage DP-inspired noise pipeline comprising latent Gaussian perturbation, IQR-calibrated Laplace noise, and 3% categorical flipping.
- **Established statistical fidelity across all 8 columns**, with 9 hypothesis tests yielding **p 0.32**, while adversarial testing produced membership-inference AUC of **0.51/0.59**, DCR ratio **0.802**, and NNAA **0.303**; grid search identified the operating point maximizing noise without sacrificing fidelity.

SKILLS

Technical Languages: C++, Python, JavaScript

Data Structures & Algorithms: Advanced Data Structures, Algorithms, OOPs

Web Development: ReactJs, Node.js, Express.js, REST APIs, HTML5, CSS3, TypeScript, Flask

Data Science: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Tableau

Databases: MongoDB, PostgreSQL, Oracle SQL

Soft Skills: Problem Solving, Leadership, Analytical Thinking, Cross-Functional Collaboration, Ownership & Accountability

Languages: English, Hindi

SCHOLASTIC ACHIEVEMENTS

- Solved 900+ DSA problems on LeetCode, achieving a contest rating of 1647.
- Solved 600+ problems on GeeksforGeeks, earning a coding score of 1786.
- Attained a 6-star rating in Problem Solving on HackerRank.
- Earned a 5-star rating in Python on HackerRank and HackerEarth.

POSITIONS OF RESPONSIBILITY

Placement Cordinator - Office of Career Servies, IIT Hyderabad